

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: AFRIDOC_MT | Context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's source content is generic English WHO articles, not Ethiopia-specific clinical material. The user accepts generic coverage as sufficient for current scope, but the deployment serves a context with Ethiopia-specific endemic conditions (kala-azar, trachoma, malaria), the Ethiopian calendar, and tiered referral pathways — all explicitly absent [\[Q103, Q105\]](#). Dataset analysis confirms a substantial share of sampled health documents cover topics geographically/contextually remote from Ethiopia [\[DATASET-D7, D32, D33, D34, D26\]](#). The benchmark itself flags translationese risk — translated Amharic sentences may exhibit unnatural syntax, overly literal phrasing, and English-centric structural and cultural framing [\[Q103, Q104, Q105\]](#), and dataset inspection confirms visible English-loanword and parenthetical-acronym patterns in Amharic [\[DATASET-D43, D47\]](#). With IC marked HIGH priority by the user, these gaps drive a score of 2: significant construct-irrelevant variance from non-Ethiopian topics combined with Ethiopia-specific construct underrepresentation.

ID	Evidence
Q103	"A potential limitation of our dataset is the influence of translationese (Koppel and Ordan, 2011). Since all source material translated originates in English, translated sentences in African languages may exhibit patterns such as unnatural syntax or overly literal phrasing." (p.10)
Q104	"Although we have not conducted an analysis to quantify these effects, prior work suggests that they can affect MT model performance, generalization and evaluation including direct assessment (Freitag et al., 2019; Edunov et al., 2020)." (p.10)
Q105	"Furthermore, AFRIDOC-MT may reflect a bias toward English in terms of structure, semantics, and cultural framing. We leave a deeper investigation of these issues to future work." (p.10)
D7	ነገር ግን፣ በ2023 እስከ ዛሬ 46 ደብሊዊፒቪ1 አዎንታዊ የአካባቢ ናሙናዎች ... ናንጋርሃር እና ኩናር አውራጃዎች ... ሚሊዮን ተመለሾች — Amharic WHO article on Afghanistan polio — geographic specificity to Afghanistan/Pakistan, not Ethiopia; pseudo-document chunk (10 sentences)
D26	ዚካ ቫይረስ ... ፀረ-ተባይ ... Vector control operations framework for Zika virus — Short Zika virus sentence fragment — topic not endemic to Ethiopia
D32	Crimean-Congo haemorrhagic fever (CCHF) is a viral haemorrhagic fever usually transmitted by ticks ... CCHF is endemic in all of Africa, the Balkans, the Middle East and in Asia. — CCHF article — hemorrhagic fever not a primary health priority in Ethiopia; tick-borne virus
D33	A volcano is a vent in the Earth's crust from which eruptions occur... There are about 1500 potentially active volcanoes worldwide. — Volcanic eruption WHO article — not relevant to Ethiopian clinical translation deployment
D34	Afghanistan continues to implement an intense campaign schedule, focusing on improving quality in the endemic zone... massive population movement significantly increases the risk of cross-border poliovirus spread — WHO Afghanistan polio article — geographically specific to Afghanistan, not Ethiopia
D43	ፀረ-ተህዋሲያን - ፀረ-ባክቴሪያ ፣ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርባግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible

Strengths

- WHO-sourced register matches the institutional voice of MOH-distributed health communication materials [\[Q18, DATASET-D11, D41\]](#)

- Maternal-child health and immunization subset (complementary feeding, breastfeeding, vaccination) is directly aligned with deployment document types [DATASET-D4, D15, D42, D44]
- Authors document data-selection ethics with attention to bias minimization and harmful-content screening [Q106, Q107]

ID	Evidence
Q18	"We scraped English articles from the websites of Techpoint Africa and the World Health Organization (WHO). The articles cover different topics of different lengths with an average length of 30 and 37 sentences for health and tech respectively." (p.3)
Q106	"AFRIDOC-MT was created with the utmost consideration for ethical standards. The English texts translated were sourced from publicly available and ethically sourced materials. The data sources were selected to represent different cultural perspectives, with a focus on minimizing any potential bias." (p.10)
Q107	"Efforts were made to ensure the dataset does not include harmful, biased, or offensive content via manual inspection." (p.10)
D4	ተጨማሪ ምግብ ማጠቃለያ፡- በ 6 ወር አካባቢ የሕፃን የኃይል ፍላጎት ... ጨቅላ ህፃናት ... ጡት ማጥባት — Amharic complementary feeding article — maternal-child health topic directly relevant to deployment (maternal-health booklets)
D11	Measures for the prevention of cholera mostly consist of providing clean water and proper sanitation to populations who do not yet have access to basic services, as well as vaccination with Oral Cholera Vaccines. — Sentence-level WHO cholera prevention content — cholera is endemic in Ethiopia but article is generic global
D15	ደህንነቱ የተጠበቀ፡- በንጽህናና ተከማቻተው እና ተዘጋጅተዋል እንዲሁም ጠርቦሶች እና ፕላስቲኮችን ሳይሆን በንጹህ እቃዎች ... ጡት ማጥባት — Complementary feeding clinical guidance in Amharic — high relevance (maternal health booklet content)
D41	WHO helps to restore primary care services so that facilities can deliver essential services, including immunization, basic treatment for common illnesses, acute malnutrition and maternal care — WHO cyclone response includes immunization + maternal care — partially relevant content
D42	የዓለም ጤና ድርጅት ሕፃናትን ለ6 ወራት ብቻ እንዲያጠቡ ይመክራል ... ደህንነቱ የተጠበቀ እና ተጨማሪ ምግብ — Amharic WHO recommendation on exclusive breastfeeding — direct match for maternal-health booklet content
D44	ጨቅላ ህፃናት ከ6 ወር ጀምሮ የተጣራ፣ የተፈጠሰ እና ከፊል ጠጣር ምግቦችን መመገብ ይችላሉ — Amharic instruction for pureed/mashed foods from 6 months — clinically precise maternal-health instruction

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: The deployment requires Ethiopia-specific knowledge: endemic diseases (malaria, TB, kala-azar, trachoma), Ethiopian calendar dates, MOH referral pathways, regional health bureau structures. None of these is represented in the benchmark — sampled documents include Afghanistan polio, CCHF, Zika, and volcanoes, none Ethiopia-endemic [DATASET-D7, D32, D26, D33]. Web search confirmed no NLP study has evaluated Amharic MT on Ethiopia-endemic disease terminology [WEB-10].

IC-2: Source documents are publicly available WHO and Techpoint articles described as ethically sourced and screened for harmful content [Q34, Q106, Q107]. Cultural alignment with Ethiopia is partial: WHO health register is broadly compatible, but Ethiopian cultural context (religious diversity, household decision-making, Ethiopian Orthodox / Muslim practices around maternal health) is not specifically reflected.

IC-3: The benchmark explicitly acknowledges potential bias toward English in structure, semantics, and cultural framing [Q105]. Dataset inspection shows English-centric patterns: parenthetical English acronyms (AMR, CCHF), English loanwords ('superbugs'), and direct transliterations of drug names (ribavirin) embedded in Amharic [DATASET-D43, D5, D6, D47]. The tech-domain content is entirely Western/Nigerian/Indian and irrelevant [DATASET-D14, D22, D38].

IC-4: INSUFFICIENT DOCUMENTATION — the benchmark does not document recruitment of Ethiopian MOH clinical reviewers to vet content for Ethiopian relevance. Translators are named but no review of source-content cultural fit is described. Would need MOH expert review of the 334 health documents against Ethiopia MOH distribution material to confirm content alignment.

IC-5: Two main content-validity harms: (1) construct underrepresentation — Ethiopia-specific clinical content (endemic diseases, calendar, referral pathways) is absent [WEB-10]; (2) construct-irrelevant variance — non-Ethiopian-relevant topics (Afghanistan polio, volcanoes, CCHF) dilute the evaluation signal [DATASET-D7, D32, D33], and translationese effects may produce English-shaped Amharic that scores well against references but reads unnaturally [Q103, DATASET-D43].

ID	Evidence
Q34	"The health data is licensed under CC BY-NCSA 3.0, while the tech data is licensed under CC BY-NC-SA 4.0." (p.4)
Q103	"A potential limitation of our dataset is the influence of translationese (Koppel and Ordan, 2011). Since all source material translated originates in English, translated sentences in African languages may exhibit patterns such as unnatural syntax or overly literal phrasing." (p.10)
Q105	"Furthermore, AFRIDOC-MT may reflect a bias toward English in terms of structure, semantics, and cultural framing. We leave a deeper investigation of these issues to future work." (p.10)
Q106	"AFRIDOC-MT was created with the utmost consideration for ethical standards. The English texts translated were sourced from publicly available and ethically sourced materials. The data sources were selected to represent different cultural perspectives, with a focus on minimizing any potential bias." (p.10)
Q107	"Efforts were made to ensure the dataset does not include harmful, biased, or offensive content via manual inspection." (p.10)
D5	ፀረ-ተሕዋስያንን መቋቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
D6	ክራይሚያ-ኮንጎ የደም መፍሰስ ትኩሳት ... CCHF ወረርሽኞች ... ከ10-40% ሞት ... ሞቃታማ አውሎ ነፋስ — Amharic article on CCHF hemorrhagic fever — clinical terminology, disease outside typical Ethiopian endemic profile
D7	ነገር ግን፣ በ2023 እስከ ዛሬ 46 ደብሊዊፒቪ1 አዎንታዊ የአካባቢ ናሙናዎች ... ናጋርሃር እና ኩናር አውራጃዎች ... ሚሊዮን ተመላሾች — Amharic WHO article on Afghanistan polio — geographic specificity to Afghanistan/Pakistan, not Ethiopia; pseudo-document chunk (10 sentences)
D14	ደቡብ አፍሪካ ጎጂ ይዘቶችን በመስመር ላይ ማቆም ትፈልጋለች ... YouTube ... NALA ... Cellulant ... ሴላኒት 20% — Long Amharic tech article: South Africa content regulation, YouTube, NALA remittances, Cellulant layoffs — entirely irrelevant to Ethiopian clinical translation
D22	Fatanmi identifies its effect on marketing a business ... Gen Zs ... Building in public... — Tech article about startup marketing strategy — entirely irrelevant to Ethiopian MOH clinical translation deployment
D26	ዚካ ቫይረስ ... ፀረ-ተባይ ... Vector control operations framework for Zika virus — Short Zika virus sentence fragment — topic not endemic to Ethiopia
D32	Crimean-Congo haemorrhagic fever (CCHF) is a viral haemorrhagic fever usually transmitted by ticks ... CCHF is endemic in all of Africa, the Balkans, the Middle East and in Asia. — CCHF article — hemorrhagic fever not a primary health priority in Ethiopia; tick-borne virus
D33	A volcano is a vent in the Earth's crust from which eruptions occur... There are about 1500 potentially active volcanoes worldwide. — Volcanic eruption WHO article — not relevant to Ethiopian clinical translation deployment
D38	የሕንድ ኢ-ቴክ ኢንዱስትሪ ... ሕንድ ከዩናይትድ ስቴትስ (ዩ ኤስ) ቀጥሎ ... edtech ... Central Square Foundation ... Nigerian Edtech ecosystem — Indian/Nigerian edtech article — entirely irrelevant to Ethiopian clinical translation
D43	ፀረ-ተሕዋሲያን - ፀረ-ባክቴሪያ፣ ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርቦግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible
WEB-10	JMIR 2025 scoping review identified only 3 Amharic NLP-for-public-health studies; none evaluated Ethiopia-endemic disease terminology (https://pmc.ncbi.nlm.nih.gov/articles/PMC11923465/)

Information Gaps

- Quantitative breakdown of how many of the 334 health documents are deployment-relevant (maternal-health / vaccination / clinical-guideline) vs. policy/financing/non-Ethiopian-endemic
- Whether the AFRIDOC-MT 'terminology harmonization' step (Q121) was actually completed and what reference materials were used

- Whether translationese effects in Amharic are systematic enough to bias automated metric scores against systems producing more natural Amharic

Requires Expert Verification

- MOH clinical-translator review of a sample of AFRIDOC-MT Amharic translations against MOH-distributed materials to assess register alignment
 - Comparison of WHO source-document language against actual Ethiopian MOH maternal-health booklets and clinical practice guidelines
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Remediation

Gap 1: WHO-sourced content includes substantial geographically remote material (Afghanistan polio, CCHF, volcanoes) and lacks Ethiopia-specific endemic-disease, calendar, and referral-pathway content; translationese effects are visible in Amharic translations.

Recommendation: Construct an Ethiopia-specific evaluation supplement of ~50–100 source documents drawn from actual MOH maternal-health booklets, vaccination schedules, and clinical practice guidelines — covering endemic diseases (malaria, TB, kala-azar, trachoma), Ethiopian calendar dates, and tiered-referral terminology. Use AFRIDOC-MT only for general-MT signal and weight Ethiopia-supplement scores higher in deployment decisions.